



STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

05 – Derivation of Component Load Combinations

This attachment describes the component load combinations (CLCs) for marine accelerations. These CLCs will be used in the code load combinations.

DNVGL section 5.6.15.1 permits the combination of peak values.

Marine acceleration loads are described using the LTV coordinate system (Figure 1).

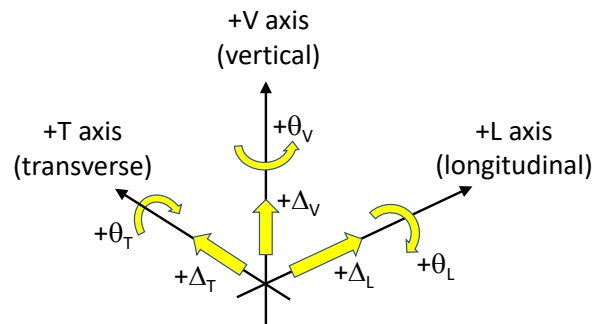


Figure 1: LTV Right-Handed Coordinate System Orientation

Component load combinations will be consolidated based on aligned horizontal forces above the origin (Figure 2).

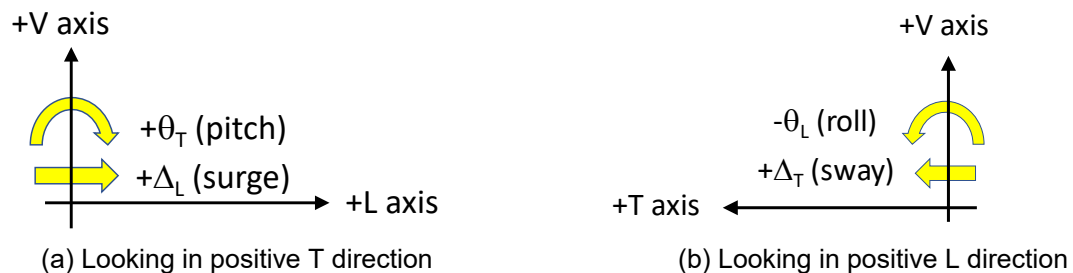


Figure 2: Aligned Rotations & Translations

In consideration of the four center of gravity shifts and four deck positions, load combinations are compiled such that CG shifts, position shifts, and lateral loading are aligned. For example, a forward-port CG shift, a forward-port position shift, and forward-port lateral loads would be applied. Between all four shifts, all loading possibilities are used.

1.0 HEAD SEA CONDITION

Included: pitch, surge, heave

Excluded: roll, yaw, sway

Included with code load combination: wind, hog/sag

The 3 included loads, each with positive and negative directions, create a total of 8 possibilities. To filter out opposing loads, use combinations where pitch and surge have the same signs (Figure C2a). Including heave, this results in 4 component load combinations.



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(CG shift & deck position = forward and port or starboard)

$H1 = + (\text{surge} + \text{pitch}) + (\text{heave})$

$H2 = + (\text{surge} + \text{pitch}) - (\text{heave})$

(CG shift & deck position = aft and port or starboard)

$H3 = - (\text{surge} + \text{pitch}) + (\text{heave})$

$H4 = - (\text{surge} + \text{pitch}) - (\text{heave})$

2.0 BEAM SEA CONDITION

Included: yaw, roll, sway, heave

Excluded: pitch, surge, hog/sag

Included with code load combination: wind

The 4 included loads, each with positive and negative directions, create a total of 16 possibilities. To filter out opposing loads, use combinations where roll and sway have opposing directions (Figure C2b). This results in 8 component load combinations.

(CG shift & deck position = port side and either fore or aft)

$B1 = + \text{yaw} + (\text{sway} - \text{roll}) + \text{heave}$

$B2 = + \text{yaw} + (\text{sway} - \text{roll}) - \text{heave}$

$B3 = - \text{yaw} + (\text{sway} - \text{roll}) + \text{heave}$

$B4 = - \text{yaw} + (\text{sway} - \text{roll}) - \text{heave}$

(CG shift & deck position = starboard side and either fore or aft)

$B5 = + \text{yaw} - (\text{sway} - \text{roll}) + \text{heave}$

$B6 = + \text{yaw} - (\text{sway} - \text{roll}) - \text{heave}$

$B7 = - \text{yaw} - (\text{sway} - \text{roll}) + \text{heave}$

$B8 = - \text{yaw} - (\text{sway} - \text{roll}) - \text{heave}$

Note: If yaw is not provided/considered, then the above list may be reduced accordingly.

3.0 QUARTERING SEA CONDITION:

Included: all translations, all rotations

Excluded: hog/sag



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Included with code load combination: wind

The 6 included loads, each with positive and negative directions, create a total of 64 possibilities. To filter out opposing loads, use combinations where pitch and surge have the same direction (Figure C2a) and where roll and sway have opposing directions (Figure C2b). This results in 16 component load combinations.

In addition, DNVGL section 11.6.4 states that the quartering condition can use 60% of longitudinal loads and 80% of transverse loads. Thus,

- Longitudinal loads (pitch, surge) are multiplied by 0.6
- Transverse loads (roll, sway) are multiplied by 0.8
- Vertical loads (yaw, heave) are not modified.

(CG shift & deck position = forward, port side)

$$Q1 = + \text{yaw} + 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q2 = + \text{yaw} + 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

$$Q3 = - \text{yaw} + 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q4 = - \text{yaw} + 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

(CG shift & deck position = forward, starboard side)

$$Q5 = + \text{yaw} + 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q6 = + \text{yaw} + 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

$$Q7 = - \text{yaw} + 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q8 = - \text{yaw} + 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

(CG shift & deck position = aft, port side)

$$Q9 = + \text{yaw} - 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q10 = + \text{yaw} - 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

$$Q11 = - \text{yaw} - 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q12 = - \text{yaw} - 0.6(\text{surge} + \text{pitch}) + 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

(CG shift & deck position = aft, starboard side)

$$Q13 = + \text{yaw} - 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) + (\text{heave})$$

$$Q14 = + \text{yaw} - 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

$$Q15 = - \text{yaw} - 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) + (\text{heave})$$



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$$Q16 = - \text{yaw} - 0.6(\text{surge} + \text{pitch}) - 0.8(\text{sway} - \text{roll}) - (\text{heave})$$

Note: If yaw is not provided/considered, then the above list may be reduced accordingly.